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— Extension of the *Best Paper Award* winner at the *4th International Symposium on Swarm Behavior and Bio-Inspired Robotics 2021 (SWARM 2021)*, Jun 1–4, 2021. Kyoto, Japan. Virtual event.
- [9] **Choi, T.** and G. Cielniak. Adaptive Selection of Informative Path Planning Strategies via Reinforcement Learning. In: *Proceedings of the 10th European Conference on Mobile Robots (ECMR 2021)*, Aug 31–Sep 3, 2021. Bonn, Germany. Virtual event. doi:10.1109/ECMR50962.2021.9568796
- [10] **Choi, T.**, B. Pyenson, J. Liebig, and T. P. Pavlic. Identification of Abnormal States in Videos of Ants Undergoing Social Phase Change. In: *Proceedings of the 35th AAAI Conference on Artificial Intelligence (AAAI-21)*, Feb 2–9, 2021. Virtual event. doi:10.1609/aaai.v35i17.17794
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- [13] **Choi, T.**, S. Kang, and T. P. Pavlic. Learning Local Behavioral Sequences to Better Infer Non-local Properties in Real Multi-robot Systems. In: *Proceedings of the 2020 IEEE International Conference on Robotics and Automation (ICRA 2020)*, May 31–June 4, 2020. Paris, France. Virtual event. doi:10.1109/ICRA40945.2020.9196728
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- INVITED TALKS [18] Self-supervised Learning of Visual Anomalies in Strawberries. In: *International Conference on Digital Technologies for Sustainable Crop Production (DIGICROP)*, Mar 2022. Virtual event.
- [19] AI Research in Agriculture and Beyond – Successful Machine Learning under Limited Resources. In: *Hankyong National University, School of Computer Engineering & Applied Mathematics Seminar*, Dec 2021. Anseong, South Korea.
- [20] Local Behavior Learning for Social Temperature Prediction without Individual Ant Tracking. In: Oral Session at *Collective Information Processing*, Mar 2020. Berlin, Germany.
- [21] Machine Learning Applications for Video Game Development. In: *Hankyong National University, School of Computer Engineering & Applied Mathematics Seminar*, Jul 2015. Anseong, South Korea.
- EXTERNAL GRANTS **Assurant**
- PI, “Help Desk Call Recording Analysis”, \$35,000, Oct 2025–Apr 2026.
- NSF**
- PI, “ERI: Odor Representation Learning for Robust Detection of Co-present Foods and Pathogen Contamination Using an Electronic Nose”, \$199,995, Jul 2025–Jun 2027.
- Georgia Peanut Commission**
- PI, “Night Owl: A Low-Cost Smart Drone System for Defending Peanut Farms from Night-time Wildlife Intrusion”, \$25,000, Jul 2025–Jun 2026.
- USDA (through SARE)**
- PI, “MoCoBot: Developing a Low-Cost Night-time Mollusk Control Robot for Strawberry Growers”, Graduate Student Grants, \$21,964, Sep 2024–Aug 2026.
- TEACHING EXPERIENCE **KSU, Marietta, GA, USA**
- Instructor*
- IT 7143: Cloud Analytics **Spring 2024, Spring 2025**
  - IT 7133: Enterprise AI Applications **Summer 2025**
  - IT 7103: Practical Data Analytics **Fall 2023, Fall 2024, Fall 2025**
  - IT 4773: Machine Learning for Enterprise **Fall 2023, Fall 2024**
- ASU, Tempe, AZ, USA**
- Teaching Assistant*
- CSE 450/551: Design Analysis of Algorithms/Foundations of Algorithms: **Jan 2018 – May 2018**
    - Instructor: Dr. Andréa Richa
    - Graded exams, and held office hours (2 hours/week) for tutoring students.

- CSE 310: Data Structures and Algorithms: **Aug 2017 – Dec 2017**
  - Instructor: Dr. Andréa Richa
  - Taught recitation sessions (4 hours/week), graded exams, and provided instructions for C++ programming assignments.
- CSE 205: Object-Oriented Programming & Data Structures **Jan 2016 – May 2016**
  - Instructor: Dr. Xuerong Feng
  - Graded exams, and tutored students for Java programming (4 hours/week).
- CSE 100: Prin. of Programming with C ++ **Jan 2016 – May 2016**
  - Instructor: Dr. Phillip Miller
  - Supervised C++ programming laboratories (5 hours/week), and held tutoring hours (4 hours/week).
- CSE 424: Capstone Project II **Aug 2015 – Dec 2015**
  - Instructor: Dr. Debra Calliss
  - Supervised each project group with their short-term and long-term goals, and graded IT ethics essays.

## MENTORING

**KSU, Marietta, GA, USA**

- Cameron Redovian (Ph.D. in Computer Science) **Aug 2024 – Present**
- Robel Mamo (M.S. in Computer Science) **Aug 2024 – Present**
- Amirmohammad Naddaf Shargh (Ph.D. in Computer Science) **Aug 2025 – Present**
- Ayushi Bharath Raj (M.S. in Information Technology) **Aug 2025 – Present**
- Ian Nortey (B.S. in Software Engineering) **Aug 2025 – Present**

**UCD, Davis, CA, USA**

- (Team 1) Satyam Goyal, Rohan Sheth, Akash Seelam, and Kavya Sasikumar  
(Team 2) Grisha Bandodkar, Athul Krishna Sugghosh, Sahil Singh, and Shyam Agarwal (B.S. in Computer Science) **Jun 2023 – Feb 2024**
  - Serving as a mentor for the EAAI-24 Mentored Undergraduate Research Challenge
  - Advising in designing a problem, producing a solution and evaluation results, and writing a paper to submit.
- Grisha Bandodkar, Zifei Cheng, Chonghan Wang, and Satyam Goyal (B.S. in Computer Science) **Jan 2023 – Aug 2022**
  - Directing to generate and extend DAVIS-Ag—a public image dataset of simulated plants—with labels, such as instance segmentation, bounding box, camera pose, etc., in a desirable format.
  - Advising in designing reasonable heuristic methods for active vision in agricultural environments and in training a fruit detector for validation of the DAVIS-Ag dataset.
  - Guiding while exploring embedded AI platforms like AI Habitat and AI2-THOR
- Avishai Halev (Ph.D. in Applied Mathematics) **Sep 2022 – Dec 2022**
  - Working with food scientists on efficient parameter search to build a realistic oxidation model fitting empirical datasets.

**UoL, Lincoln, UK**

- Owen Would (M.Sc. in Robotics & Autonomous Systems) **Mar 2021 – Sep 2021**
  - Advised in studying on deep neural network-based visual anomaly detection of strawberry images particularly while designing a GAN-based method for occluding environments.

**ASU, Tempe, AZ, USA**

- Sehyeok Kang (M.S. in Computer Engineering) **Mar 2019 – May 2020**
  - Mentored to implement Remote Teammate Localization on a physical robot platform, *Thymio*, while he was working on his thesis on correlation between prediction accuracy

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|                      | and observational information.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |
|                      | <ul style="list-style-type: none"> <li>• Ricardo Weir (B.S. in Computer Science) <b>Mar 2018 – Dec 2018</b> <ul style="list-style-type: none"> <li>– Guided to develop a YOLO-based object-detection pipeline—from data annotation to model validation—to perform automated tracking of individual <i>Harpegnathos</i> ants from high-quality video recordings.</li> </ul> </li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |  |
| MEDIA COVERAGE       | <ul style="list-style-type: none"> <li>• Semafor (Sep 2025) – “It’s common scents”</li> <li>• WSB Radio (Sep 2025) – “Kennesaw State researchers develop ‘Electronic Nose’ to detect food-borne illness”</li> <li>• Fox 5 Atlanta (Sep 2025) – “KSU researcher developing AI-powered ‘electronic nose’”</li> <li>• The Georgia Sun (Sep 2025) – “A KSU Researcher is Creating an Electronic Nose to Sniff Out Foodborne Illness”</li> <li>• KSU News (Sep 2025) – “KSU researcher developing electronic nose to sniff out foodborne illness”</li> <li>• KSU News (Jun 2025) – “Kennesaw State researchers developing AI-powered drones to protect peanut farms from wildlife”</li> <li>• Wisconsin Ag Connection (May 2025) – “Night Vision Robot Protects Strawberry Crops”</li> <li>• WSB Radio (May 2025) – “Kennesaw State University researchers develop AI robot for pest control”</li> <li>• KSU News (May 2025) – “Kennesaw State researcher develops AI robot to aid farmers in fight against pests”</li> </ul>                                                                                                                                                                                          |  |
| PROFESSIONAL SERVICE | <p><b>KSU Research Champion</b> <b>Aug 2025 – Present</b></p> <ul style="list-style-type: none"> <li>• Co-leading the Applied Technologies Research Community, promoting collaborative research across the campus.</li> </ul> <p><b>Workshop Organizer</b></p> <ul style="list-style-type: none"> <li>• ICRA2023: “TIG-IV: Agri-Food Robotics From Farm To Fork” <b>Jun 2023</b> <ul style="list-style-type: none"> <li>– Full-day workshop on robotic innovations for agri-food systems—from farming to post-harvest processing, cooking, delivery, serving, and legislation</li> </ul> </li> </ul> <p><b>Conference/Journal Reviewer</b></p> <ul style="list-style-type: none"> <li>• IJRR, RAM, RA-L, Biosystm. Eng., Comput. Electron Agric., IROS 2025 (Associate Editor), IROS 2023, ICRA 2023, ICRA 2022, IROS 2022, ICRA 2020</li> </ul> <p><b>Grant Reviewer</b></p> <ul style="list-style-type: none"> <li>• Research Grant at ASU GPSA <b>Aug 2017 – May 2018</b></li> <li>• Travel Grant at ASU GPSA <b>Aug 2016 – Jul 2017</b></li> </ul> <p><b>Conference Session Chair</b></p> <ul style="list-style-type: none"> <li>• TAROS 2021 <b>Sep 2021</b></li> <li>• CASE 2017 <b>Aug 2017</b></li> </ul> |  |
| AWARDS               | <p><b>KSU</b></p> <ul style="list-style-type: none"> <li>• Apex Award for research and scholarly excellence <b>Dec 2025</b></li> </ul> <p><b>KSU PechaKucha Night</b></p> <ul style="list-style-type: none"> <li>• Best Demonstration Award <b>Nov 2024</b></li> </ul> <p><b>IROS2023 Workshop on Agricultural Robotics for a Sustainable Future</b></p> <ul style="list-style-type: none"> <li>• 2nd Prize for Innovation in Precision Agriculture <b>Oct 2023</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |

**SWARM 2021**

- Best Paper Award

**Jun 2021****ASU Graduate College**

- Completion Fellowship (\$8,550 plus tuition for 1 credit hour)

**Aug 2020****ASU Ira A. Fulton Schools of Engineering**

- Engineering Graduate Fellowship (\$700)

**May 2020****ASU School of Computing, Informatics, and Decision Systems Engineering**

- Doctoral Fellowship (\$4,000)

**Mar 2020****ASU Social Insect Research Group**

- Student Research Grants (\$1,550)
  - Project: Deep Features for Generalizable Insect-behavior Learning.

**Nov 2018****SSU College of Information Technology**

- Bronze Award at Software Development Competition
  - Social Alarm: Smart Android Alarm Application (Photos & Demo)

**Oct 2012****WORK  
EXPERIENCE****Atlassian**, Mountain View, CA*Data Scientist Intern***May 2018 – Aug 2018**

- Jira Duplicate Ticket Detection
  - Designed a deep learning pipeline for human natural language to classify semantically similar tickets from customers.
  - Gathered >124K examples to implement, train, fine-tune, and validate specialized LSTM models.
  - Demonstrated 1) significantly higher accuracy than traditional machine learning models, 2) generalizability to the data from different sources of ticket, and 3) feasibility of similarity-based ranking scenarios.

**HARDWARE AND  
SOFTWARE SKILLS****Data Science & Machine Learning:**

- PyTorch, TensorFlow, TensorBoard, OpenAI Gym, OpenCV, Scikit-learn, SciPy, NumPy, Pandas, Matplotlib, etc.

**Programming Languages:**

- Python, Java, C, C++, UNIX shell scripting, GNU make, MySQL, etc.

**Operating Systems:**

- Microsoft Windows family, Apple OS X, Linux, and other UNIX variants

**Others:**

- Unity 3D, MATLAB,  $\text{\LaTeX}$ , Git, Android, and TCP/IP programming